Chapter 13

Impression materials and techniques

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INTRODUCTION

Impressions of teeth and related structures are taken for making implants, crowns, bridges and dentures, and the material selected will partly depend on the personal choice of the clinician and the use. The choice of impression tray used will also be dictated by the impression material and its viscosity. Impression materials used for advanced operative procedures such as crown and bridgework must have elastic properties to allow the removal from undercut areas in the mouth without creating permanent distortion; these can be either water based (hydrocolloids) or synthetic elastomers.

THE MATERIALS

Hydrocolloid impression materials

Agar is an example of a *reversible hydrocolloid* impression material. It consists of long chain polysaccharide molecules and water. When cool, the polysaccharide chains are linked together by hydrogen bonds between hydroxyl groups on the polysaccharide chains, resulting in the characteristic gel. When the temperature is raised, the hydrogen bonds are broken and the gel liquefies; on cooling again the gel state reforms, hence the name 'reversible hydrocolloid'. Whilst it is a hydrophilic material and gives an accurate impression, it suffers from a number of disadvantages, namely it is dimensionally unstable and needs to be cast up within 1 hour, it has low tear resistance, is hot and uncomfortable on insertion and requires additional equipment such as

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specially designed water baths, impression trays and tubing (for the water coolant). The use of a water bath is a major drawback in relation to contamination and problems with cross infection control; as such, the material is rarely used in contemporary practice.

Alginate is an *irreversible hydrocolloid* material whose main ingredients are sodium alginate, calcium sulphate dihydrate and inert filler particles. Alginates are long polysaccharide chains derived from a marine plant (seaweed). It is presented as a powder which is mixed with water. When this is done a setting reaction takes place which involves the calcium ions irreversibly cross-linking the polysaccharide chains, so converting the sol into a gel. This would be a rapid reaction if a retardant was not added. Sodium phosphate is therefore added which preferentially reacts with the released calcium ions forming calcium phosphate; the calcium ions initially are therefore not available to cross-link the polymer chains. Once all of the sodium phosphate has been used up in this reaction, the remaining calcium released is used in forming the cross-linkages and the gel. The filler particles give the set material its rigidity.

Alginate is a commonly used material but suffers from the same disadvantages of the reversible hydrocolloids, that is, they have poor tear resistance and are dimensional unstable. If alginate impressions are stored dry, they dry and shrink; if stored under water, they imbibe water (imbibition) and swell. In addition, the impression is not as detailed as that taken with a synthetic elastomer and therefore not suitable for impressions of crown and bridge preparations; its main use is for study casts and opposing models.

Before taking the impression an appropriately sized impression tray should be selected (Figure 13.1). A perforated plastic stock tray is ideal for this as the material is not too viscous and should not distort the tray on seating. Such trays are for single use only and would distort if heated during a decontamination process. Although a metal tray could be used as an alternative (Figure 13.2), thorough cleaning and decontamination of the trays between patients is time consuming and, as such, disposable single-use plastic trays are more popular and cost effective. A tray should be selected that seats fully over the dentate arch with its flanges not bedding into the soft tissues of the buccal, labial and lingual sulci. Occasionally, the stock tray needs modifying with impression compound to increase its extension to completely cover the dentition and areas of interest.



Figure 13.2 Metal non-perforated impression stock trays. The rolled metal rim affords some degree of retention to the set impression material; however, an adhesive is also required.

Despite the fact that stock trays have perforations and ridges to retain the set impression material in the tray, a tray adhesive should also be used. Alginate adhesive is supplied either in a bottle for use with a brush or in an aerosol can (Figure 13.3). Brushing the adhesive on the fit surface of the tray and just over its flanges ensures an



Figure 13.1 Selection of impression stock trays. Light blue plastic (opaque) trays are suitable for alginate impressions. Clear coloured polycarbonate trays are more rigid and are suitable for more viscous elastomeric impression materials. Three colour-coded sizes are seen.



Figure 13.3 Selection of tray adhesives. The alginate adhesive is provided as either paint on (top right) or aerosol (bottom). A universal adhesive can be used for silicone impression materials (top left) and a specific polyether adhesive (top centre) should be used for polyethers such as Impregum (3M ESPE).

even thin layer of adhesive; however, the adhesive needs to be dispensed into a disposable dappens pot first and applied with a single-use brush to prevent contamination of the adhesive in the bottle. To simplify matters and avoid problems with cross-infection control, aerosol adhesives can be used. The work surface should be covered with a paper napkin and the fit surface sprayed with an even thin application of adhesive. Before use the adhesive should be dry and tacky; if the adhesive is applied too thickly it will not dry and will act as a lubricant, with the risk of the impression material pulling away from the tray on withdrawal from the mouth.

When mixing alginate it is important to use the correct proportion of powder to water. To achieve this, manufacturers provide a measuring scoop for the powder and a measuring cylinder for the water. Before use the powder should be stirred in its container to thoroughly mix the ingredients and to ensure the material has not become compacted during storage. The number of scoops of powder will be dictated by the size of the impression tray and the patient's mouth. The powder is mixed with the correct amount of water and evenly combined in a mixing bowl. Most alginate materials have a mixing time of about 1 minute. The mixed alginate is then loaded into the impression tray to the level of the flanges and posterior extremity using a spatula. Overloading the tray should be avoided as this will lead to difficulty in seating the impression in the mouth and will be unpleasant for the patient, with a risk of obstructing the airway.

When taking alginate impressions the patient should be seated upright in the dental chair with their head at about the elbow height of the operator. Once the tray has been loaded with alginate, a portion of alginate can be scooped up on the operator's finger and smeared over the occlusal surfaces of the teeth and any areas of importance. Rubbing the material into fine recesses such as pits and fissures reduces the risk of incorporating any air bubbles. The loaded impression tray is then seated before the smeared alginate starts to set (single-stage technique). For the lower impression, if right handed, the operator should stand on the patient's right-hand side facing the patient. The impression tray is then rotated into the mouth, with that part of the tray on the patient's left entering first. Once the tray is over the dentate arch the lips should be retracted before seating the tray, allowing the alginate material to flow into the sulci. The patient should then be asked to raise the tongue over the tray to mould the lingual sulcus and the material in the buccal and labial sulci can be moulded with external light pressure on cheeks and lips. If the operator is left handed they should stand on the patient's left-hand side and carry out the procedure in a similar mirror fashion.

The upper impression is taken in a similar way, with the exception that the operator now stands behind the patient's right shoulder (if right handed) and the tray is rotated into the mouth with the patient's right side of tray first. Once positioned over the dental arch, the tray should be seated posteriorly first, lifting the lip up and out when seating the tray anteriorly. This ensures that the impression material flows forwards and not backwards towards the patient's throat. If the patient retches it is important not to remove the tray when the material is unset, but ask the patient to lean forwards and breathe deeply through their nose. This should effectively close the oral cavity from the oropharynx and prevent material flowing posteriorly.

The speed with which the material sets can be altered by the manufacturer, and regular and rapid-set materials are available with working and setting times ranging from 1 to 4 minutes depending on the material chosen. The setting time can also be influenced by the temperature of the water; using warmer water will speed up the setting reaction. Alginate when initially mixed is alkaline (pH $\approx$11) but when set is near neutral. Some manufacturers have taken advantage of this and include a pH indicator which changes colour with the pH. Once set, the tray should be removed from the mouth with a rapid technique. This is because the material is viscoelastic and if removed rapidly elastic recovery should take place; slow removal from undercuts would lead to distortion of the material.

Once removed from the mouth any saliva should be rinsed off the surface and the impression should be decontaminated; prolonged immersion in a disinfectant solution leads to distortion of irreversible hydrocolloid impression materials. Ideally, if there is a laboratory facility, the impression should be cast within 30–60 minutes. If not, the impression should be stored at 100% humidity by wrapping the impression in wet gauze or a napkin and placing it in a sealable plastic bag during transit to the laboratory.

The impression provides the technician with a negative mould into which stone can be poured. This is done by holding the impression over a vibrator so that air bubbles in the stone rise to the surface as the material flows into all recesses. Once the impression is filled to the level of the sulcus, it is inverted and based onto stone mounded up on the bench top. Once the stone has set the impression can be removed and the excess stone trimmed. The alginate impression should not be left on the cast for any period after the stone has set as it dries out and becomes difficult to remove.

Elastomeric impression materials

Impressions for indirect restorations need to be accurate to record fine detail of tooth preparations and they need to be dimensionally stable with time. Usually there is a delay in taking the impressions and casting them up as most practices do not have a laboratory on site and have to send impressions through the post. An ideal impression material should also have good elastic recovery

when it is used in undercut situations and should have good tear resistance. The synthetic elastomeric materials in general meet these high demands. The elastomeric impression materials available are silicones (addition and condensation cured) and polyethers. Historically, *polysulphide* impression materials were included in the list of elastomeric impression materials and favoured by some for their long setting time; however, they are rarely used in contemporary practice due to a list of disadvantages which include an unpleasant smell, shrinkage on storage resulting in distorted dies, and the need to use it in conjunction with a laboratory-made special tray.

Silicones

Addition cured silicones

The most commonly used elastomeric impression material is the *addition cured silicone*. These materials consist of polyvinyl siloxane polymer chains which have terminal vinyl groups that are cross-linked by a silanol (present in the base paste) in the presence of a platinum catalyst (present in the catalyst paste). No by-products are produced during the setting reaction and, as such, the material is dimensionally stable.

Manufacturers usually produce addition cured silicone base and catalyst in putty, heavy bodied, medium bodied and light bodied forms dependent on the amount of filler content. Base and catalyst with the same viscosities are mixed together. Usually addition cured silicones are used in a two-phase, single-stage technique using a stock tray.

A two-phase, single-stage technique

The *two-phase* (or twin mix) refers to the fact that two viscosity materials are used to take a single impression, with a light body wash being syringed around the tooth preparation to record the fine detail and either a heavy body or putty material loaded into the tray to take up the bulk of space between the preparation and the tray. The wash is syringed around the tooth preparation whilst the dental nurse is mixing the heavy body or putty and loading it into the stock tray (Figure 13.4); the heavier bodied materials have the benefit that they do not flow from the tray. Before the wash begins to set, the loaded tray is seated over the teeth, with the heavier bodied material forcing the low viscosity wash into the intricate recesses around the tooth preparation (Figure 13.5). Both wash and heavier bodied materials set at the same time, hence the term *single-stage technique*.

Most light bodied materials are now supplied in dual cartridges (base and catalyst) which are loaded into guns onto which double helix mixer tips are attached (Figure 13.6). As the gun is activated, the plungers express the base and catalyst through the mixer tip which ensures a perfectly proportioned, bubble-free mix of base and catalyst. A small amount of material should be syringed onto a pad or bracket table to allow its setting to be monitored. Occasionally, especially with new cartridges, air inclusions at the end of either the base or catalyst cartridge mean that only one or other of the materials exits the mixer tip in an unmixed state which will obviously not set. Syringing the material onto the bracket table first ensures only mixed material is syringed around the teeth.



Figure 13.4 Light bodied wash being syringed around upper anterior tooth preparations.



Figure 13.5 Putty in stock tray being seated over wash before it starts to set.



Figure 13.7 Air blown onto light bodied wash, dispersing it into recesses of the preparation.

The fine tips that attach to the double helix mixers allow the mixed material to be deposited at the preparation margins and internal recesses such as slots and grooves (see Figure 13.4). Some clinicians advocate gently blowing air through the three-in-one syringe to disperse the wash and to blow it into areas such as the gingival sulcus (Figure 13.7). Care should be taken if this is done as too strong an air pressure could lead to air being incorporated into the wash with resultant bubbles in the set impression or the wash could be simply blown off the prepared tooth. The latter is more likely if there is moisture contamination on the tooth surface.

Putty materials are usually provided in tubs with colour-coded scoops for colour-coded base and catalysts, respectively; cross contamination of scoops should be avoided. Equal proportions of the base and catalyst are mixed by hand until an even, streak-free mix is obtained. As the set of some addition cured silicone materials is inhibited by residues from latex gloves, the materials should be mixed with vinyl or nitrile gloves donned; most manufacturers are overcoming this problem but if in doubt avoid latex gloves. Some putty and heavy bodied materials are provided in cartridges for use in an electrically driven Pentamix automatic mixing unit (3M ESPE) with a double helix mixer tip, giving all of the advantages that the corresponding tips have for mixing the light bodied materials (Figure 13.8).

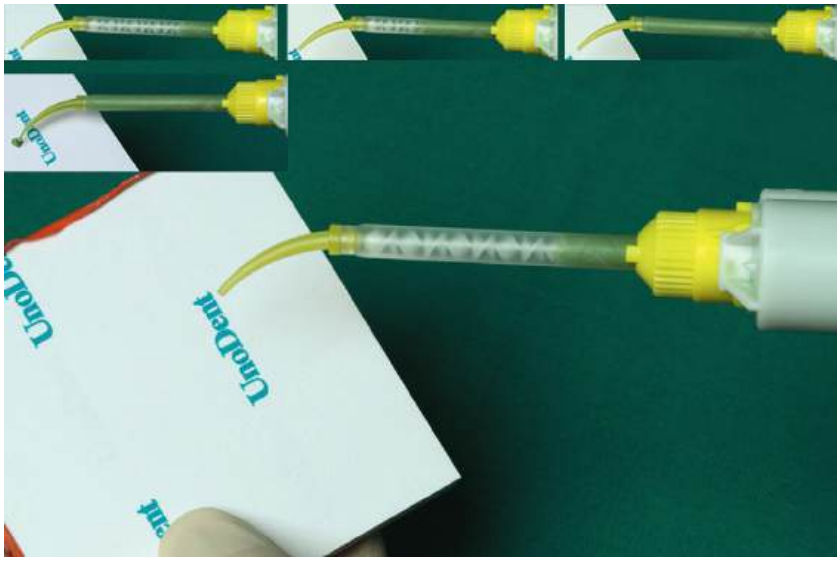


Figure 13.6 Light bodied wash cartridge containing separate base and catalyst loaded into a mixing gun to which is attached a double helix mixer tip.

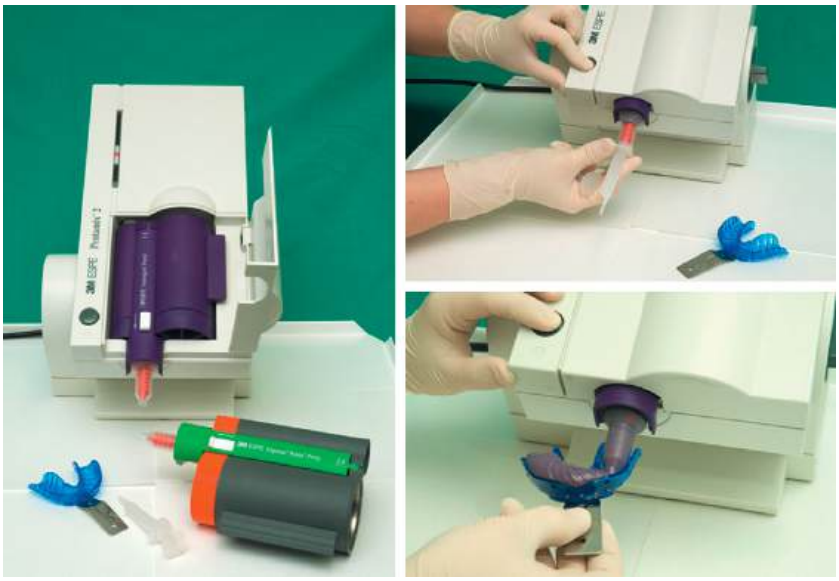


Figure 13.8 Pentamix for automated mixing of polyethers and heavy bodied impression materials (left). Polyether being loaded into a syringe (top right) and the impression tray (bottom right).

A rigid plastic or polycarbonate tray is required for such an impression technique as the heavier bodied materials could distort less rigid trays on seating (see Figure 13.1). Metal trays could be used but with caution; if the materials set into deep undercuts, removal might be difficult and in this situation sectioning the tray to salvage the situation would be almost impossible. Despite the fact that most trays have perforations and ridges to retain the impression material, an appropriate adhesive for addition cured silicones must be used (see Figure 13.3).

Two-phase, two-stage technique

As previously described, the two-phase technique refers to the fact that two viscosity materials are used, usually putty or heavy body and wash. However, the two-stage aspect refers to the fact that, in the first stage, the putty or heavy bodied material is seated in the mouth, allowed to set and then removed. The second stage to the impression involves syringing the wash around the tooth preparation(s) and, before it sets, reseating the putty or heavy bodied impression over it. This forces the wash into all the recesses of the preparation. The wash, however, takes up space and on reseating the putty impression, compression of the putty occurs (Figure 13.9). When the impression is removed the putty recoils and the distorted impression will result in a smaller die of a full coverage crown preparation, for example. The risk of distortion using this technique can be reduced in two ways which provide space for the wash between the tooth preparation and primary impression:

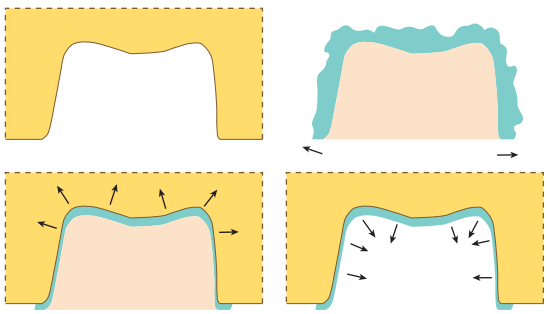


Figure 13.9 Diagrammatic representation of a two-phase, two-stage impression technique. First the putty or heavy bodied impression is taken and allowed to set (top left). Wash is syringed around the tooth preparation (top right) and before it is set the first stage impression is reseated. As this is a good 'fit' there is little space for the wash to escape and compression of the first stage impression occurs (bottom left). On removing the impression, the compressed impression recoils and provides an imprint (impression) of the prepared tooth which is smaller (bottom right). A crown made on this model will be too small and will not fit the prepared tooth.

- The first or primary impression can be taken before the tooth is prepared, resulting in greater space for the wash once tooth preparation has been carried out. However, gauging how much wash to use to completely fill this space without leaving voids is difficult.

- Use of a cellophane spacer over the tooth preparation provides a small space but distortion of the primary impression is still a high risk.

To reduce the risk of distortion further, V-shaped grooves can be cut into the primary impression that corresponds to the area between the tooth preparation and the buccal (and where possible lingual) sulcus. This effectively allows venting of the wash without excessive build-up of pressure and distortion of the putty.

Wherever possible the two-stage technique should be avoided with addition cured silicones. In addition to the problems associated with distortion of the impression, reseating rigid putty materials may be difficult, especially where there are deep undercuts.

Condensation cured silicones

Condensation cured silicones also consist of polyvinyl siloxane polymer chains and filler particles in a base paste; however, in these materials, the polymer chains end with hydroxyl terminal groups. The catalyst or activator paste contains tetraethyl silicate molecules which can cross-link up to three polyvinyl siloxane polymer chains at a time. This cross-linkage setting reaction is accompanied by the release of alcohol groups as a by-product. The release of alcohol during the setting reaction also makes the material less dimensionally stable, with the greatest shrinkage with time of all the elastomeric materials. As with addition cured materials, manufacturers produce various viscosity base pastes, but the activator paste is often of a different viscosity from the heavier bodied materials, making mixing more difficult.

Condensation cured silicones can be used in either a single-stage impression technique or a two-stage technique. In the two-stage technique any compression of the primary putty impression on reseating over the wash during the secondary impression will be compensated for to some degree by the shrinkage of the material. Compression and shrinkage cannot, however, be accurately predicted and the condensation cured materials are therefore not as accurate as the addition cured materials.

Polyethers

Polyethers are presented as a base and activator paste of the same viscosity. Unlike the silicone impression materials, polyethers are only available in one viscosity and are therefore an example of a monophase impression material; the same viscosity material is syringed around the tooth preparation(s) as is loaded into the impression tray. It is therefore a monophase material that should always be used in a single-stage technique. Polyethers have good elastic properties and are particularly suited for implant prosthodontics.

The base paste of a polyether contains inert fillers and polyether polymer chains which terminate in an amine group. These polymer chains are cross-linked by an aromatic sulphonate ester in the activator paste, which also contains inert filler particles. There are no by-products to this setting reaction and, as such, the material is dimensionally stable; however, since the materials will take up water, the set impression should be stored dry. Polyethers which are presented in tubes for manual mixing on a pad are messy and difficult to mix and it is for this impression material that the Pentamix was originally designed (see Figure 13.8). Polyether impressions should be cast within a few days as prolonged storage leads to deterioration of the material.

GENERAL CONSIDERATIONS FOR ELASTOMERIC IMPRESSION MATERIALS

Blocking out undercuts

The most commonly used elastomeric impression materials are the addition cured silicones and polyethers and these materials are the most rigid once set. It is therefore important prior to taking any impression in these materials to check intraorally for undercuts into which the materials may flow and become lodged once set. Common undercuts to look out for are buccal undercuts in the sulcus, periodontal bone loss and large interdental spaces and recesses beneath bridge pontics (Figure 13.10).



Figure 13.10 An elastomeric impression material could flow into the recess beneath the bridge pontics and make removal of the impression difficult once set as they are relatively rigid and have a high tear resistance. Such recesses (undercuts) can be blocked out with soft wax.

Undercuts can simply be blocked out intraorally by placing soft wax, ensuring the wax does not interfere with the occlusion and appearance in the aesthetic zone.

Protecting the dentine

Preparation of teeth for indirect ceramic or composite restorations, in particular full coverage crowns, leads to freshly cut dentinal tubules. These tooth-coloured restorations should be cemented with resin luting cements and a systematic review of the literature has suggested that the dentine could be sealed with a dentine bonding agent prior to taking an impression. This so-called immediate dentine seal (IDS) protects the pulp during the drying and impression taking, leads to improved bond strengths when the indirect restoration is cemented, and reduced microleakage and postoperative dentine hypersensitivity (Magne, 2005). Whether these findings can be extrapolated to metal-based restorations will depend on the subsequent luting cement used; if it is resin based, similar findings might be expected.

Keeping a dry environment

All elastomeric materials suffer from the major drawback of being hydrophobic. The degree to which a material is hydrophobic or hydrophilic is reflected in its ability to wet a surface; this is measured by the contact angle of water on the material's surface. Polyethers are less hydrophobic (contact angle 49°) than addition cured silicones (contact angle 98°). Some manufacturers have tried to overcome the problem of the highly hydrophobic silicones by adding surfactants to their materials; however, no elastomeric material can be regarded as hydrophilic and impression-taking techniques demand that the teeth are perfectly dry.

Whilst it is important to take alginate impressions (especially the upper) with the patient sitting in the upright position, most elastomeric materials are viscous or do not flow unless placed under pressure. This, together with the fact that the teeth need to be dry, necessitates that the impressions are taken with the patient lying back. It is only then that the teeth in question can be adequately dried with a three-in-one syringe and isolated with cotton wool rolls or absorbent pads and a saliva ejector placed to remove excess saliva. Once the wash has been syringed around the preparation, the cotton wool pads and saliva ejector have to be removed quickly prior to seating the loaded impression tray. Cotton wool embedded into the impression prevents appropriate decontamination prior to sending to the laboratory.

Checking the impression

When using the two-phase, single-stage impression technique, speed is of the essence. The wash should be in an unset state when the putty is seated. Once the impression material has set it is important to evaluate the impression for accuracy and any potential faults in relation to the prepared teeth:

- The wash should have merged imperceptibly with the putty (Figure 13.11). If there is a demarcated line between the wash and the putty, this indicates that the wash had started to set before the putty was seated (Figure 13.12).The impression must be retaken as the set wash is likely to have distorted under the pressure of seating the putty.
- All preparation margins should be clearly visible (Figure 13.11), with no drags or air bubbles. Taking an impression of subgingival margins can be difficult.
- No drags or defects between the putty and wash should be found corresponding to the axial walls of the preparation, internal line angles or recesses such as slots, grooves and boxes.

Focus has been placed on taking an impression of the tooth or teeth that have been prepared for indirect restorations; however, it is important to ensure that an adequate impression of the entire arch is made to ensure that the occlusion and shape of the remaining teeth can be recorded. Once the wash has been syringed around the prepared teeth, it can then be syringed into the pits and fissures of posterior teeth or areas of particular interest. Alternatively, where two-phase, single-stage impressions are concerned, wash can be syringed into the loaded impression tray corresponding to the arch form.

The impression should also be checked for errors elsewhere and areas remote from the site of the prepared tooth. This is important to confirm that a model can be produced, not only to ensure that a restoration can be made on the master die, but also to ensure that the occlusion can be formed correctly. For example, an impression might be taken of a lower molar tooth for a gold crown. Although the impression might be satisfactory for construction of the crown margins, a drag labial to the lower incisors might prevent upper and lower models seating together in a Class I incisor relationship and hence inability to check the occlusion. Occasionally the impression will pull away from the impression tray, distorting the occlusal plane. In these situations the impression should be retaken.

GINGIVAL MANAGEMENT – RETRACTION

Chapter 2 emphasized the importance of establishing periodontal health prior to the placement of indirect restorations. If this is achieved and careful tooth preparation is carried out with appropriate placement of cervical margins, there is little need for gingival management or retraction prior to taking an impression. Wherever possible preparation margins should be placed supragingivally;



Figure 13.11 An addition cured silicone putty (light blue) and wash (dark blue) impression of five upper anterior teeth. Failed porcelain jacket crowns have been removed from four of the teeth, hence the shoulder preparations. The upper right lateral incisor has a minimal veneer preparation. All margins are visible in the impression, the wash has merged with the putty and no bubbles, drags or defects are visible. The impression tray is visible in two areas, but this is unlikely to have any impact on the model cast.



Figure 13.12 An impression for a resin retained cantilever bridge using the premolar tooth as an abutment. The impression of the mesial and distal rest seats are clear; however, the wash (orange) is demarcated from the putty (green), indicating that the wash had set before the putty was seated. The impression should be retaken.

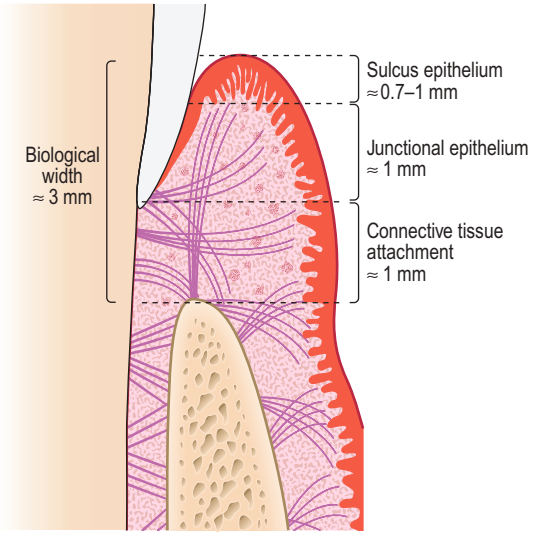


Figure 13.13 Diagrammatic representation of the biological width.

the closer the margins are to the gingivae, the greater will be the potential adverse effect on gingival health.

When considering indirect restoration margins in relation to gingival health it is important to reconsider the biological width. The supra-alveolar gingival apparatus consists of the gingival sulcus (0.7–1 mm), the junctional epithelium (≈1 mm) and the connective tissue attachment (≈1 mm), making up approximately 2.7–3 mm of biological width (Figure 13.13). The biological width makes up the body's natural defence mechanism and when



Figure 13.14 This patient has severe tetracycline staining. Original placement of the crown margins subgingivally to mask the discolouration encroached upon the biological width and led to attachment loss and recession. Poor oral hygiene has probably exacerbated the situation. Following successful basic periodontal treatment, further crowns will be required due to the patient's high lip line.

this is encroached upon, for example by a subgingival restoration, the body's reaction is to re-establish the biological width. This takes the form of alveolar bone loss, new connective tissue attachment, apical migration of the junctional epithelium and gingival recession, the crown margins now becoming supragingival (Figure 13.14).

Where appearance is of paramount importance in the aesthetic zone in patients with a high lip line, subgingival margins may have to be accepted as a compromise, but these should not extend deeper than 0.5 mm from the free gingival margin. If deep extensive restorations or short clinical crowns dictate that the margins should be more apically placed than this, surgical crown lengthening should be considered.

Occasionally a pragmatic approach has to be taken when a restoration margin extends subgingivally beyond the ideal 0.5 mm and crown lengthening is neither desirable nor appropriate. In these situations retraction of the gingiva has to be achieved in order to obtain an accurate impression of the tooth preparation margin and occasionally arrest haemorrhage. This can be achieved by mechanical retraction, chemomechanical retraction or surgical retraction. In any case a sulcus width of 0.2 mm or more is required to record the preparation margin accurately and have an adequate thickness of impression material for durability.

Mechanical retraction

Retraction cord

The most time-honoured method of mechanically retracting the gingival tissue is the use of retraction cord. Retraction cords can be twisted like a rope, braided or knitted, and are provided in different diameters (Figure 13.15). Plain retraction cord is placed into the gingival sulcus with a gentle packing technique using either a flat plastic instrument or a specially designed instrument which has a rounded, serrated head. A single length of retraction cord is usually packed into the sulcus for the full circumference of the tooth. Care must be taken not to use excessive pressure on packing retraction cord so as not to cause



Figure 13.15 Knitted retraction cord in pots of various sizes (left). Magnified image of cord (top right) and placed in situ (bottom right).

trauma to the soft tissues, minimizing subsequent recession on healing.

Twisted retraction cord tends to unwind on packing and for this reason braided cords are easier to use. When retraction cord is used in its plain, untreated form to physically push the tissues away from the tooth, gingival crevicular fluid and/or blood becomes absorbed into it. Subsequent removal of the cord can cause trauma of the sulcus epithelium or can remove any blood clot, leading to haemorrhage and problems with impression taking. Wetting the retraction cord prior to removal and re-drying the preparation may reduce the risk of this happening.

Some clinicians may argue that the retraction cord should remain in situ whilst the impression is being taken, and some even advocate a double cord technique. In the latter, a narrow first cord is packed into the gingival sulcus and a second wider cord packed on top. Only the second cord is removed prior to taking the impression, leaving the first in situ during the impression to maintain the gingival retraction. If retraction cord is left in situ it is important to ensure that it lies beyond the preparation margin so that the impression material records the entire margin and just beyond. Retraction cord lying level with the preparation margin makes it impossible to record and see the margin on the subsequent model (Figure 13.16).



Figure 13.16 Retraction cord left in situ for impression taking (left) has led to an inaccurate impression and subsequent model (right). The retraction cord sat at the level of the crown preparation margin and, as a result, the crown preparation margins cannot be seen on impression or model.

Magic Foam Cord (Coltène Whaledent)

Possibly a less traumatic method to retract gingival tissues mechanically is with the use of Magic Foam Cord (Coltène Whaledent Ltd, Sussex, UK). This material is an expanding polyvinyl siloxane which is syringed around the cervical margin of the tooth (Figure 13.17). A Comprecap, similar to a contoured cotton roll, is then held in place over the silicone. As the silicone begins to set it starts to effervesce, and it is the production of air bubbles that causes the material to expand and mechanically push the gingivae away from the tooth. The material should be left in place for up to 5 minutes and removed prior to taking the impression. It is therefore time consuming to use and does not have a haemostatic agent which limits its use.

Chemomechanical retraction

Retraction cords revisited

Astringents or vasoconstrictor-impregnated retraction cords can overcome some of the problems associated with impression taking. Plain cords can also be dipped in these

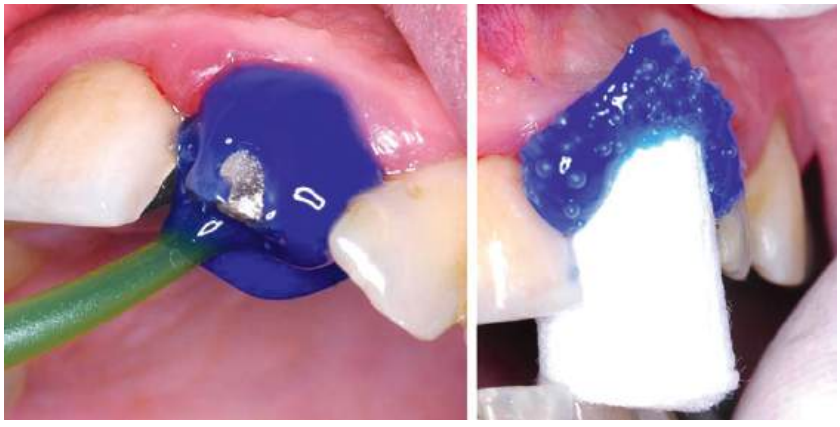


Figure 13.17 Magic Foam Cord (Coltène Whaledent) is a polyvinyl siloxane which is syringed around the tooth preparation (left). A Comprecap is then seated over the tooth (right) and on setting the silicone effervesces and expands, leading to less traumatic mechanical retraction of the gingiva.

chemicals. The cord mechanically retracts the gingivae and the chemicals help to arrest haemorrhage. Astringents such as aluminium chloride, ferric sulphate, potassium aluminium sulphate and zinc chloride help to reduce moisture, improving the impression result. The least damaging to the soft tissues and the astringent of choice is aluminium chloride.

Adrenaline has also been impregnated into retraction cords to produce vasoconstriction and arrest of haemorrhage. However, concerns have been raised over the absorption of adrenaline into the bloodstream and the concomitant systemic effects such as tachycardia, increased blood pressure and heightened anxiety, and interactions with drugs such as tricyclic antidepressants. The amount of adrenaline absorbed is unpredictable and will depend on gingival health, dilution from saliva or gingival crevicular fluid, and concentration of adrenaline used. The use of adrenaline-impregnated cord should therefore be avoided or used with caution, taking the patient's medical history into consideration.

Expasyl (Kerr Dental)

Expasyl is a relatively new method of chemomechanically retracting gingival tissues. It consists of a kaolin-based paste which is injected, using a unique gun, into the gingival sulcus (Figure 13.18). The material is very viscous and maintains its rigidity, effectively displacing the gingivae away from the tooth without the trauma that can be caused when packing retraction cord. Expasyl also contains aluminium chloride to arrest bleeding. The paste is left in place for up to 2 minutes and is then removed atraumatically by rinsing it away with a three-in-one syringe. The preparation then needs to be dried thoroughly prior to taking the impression.

Surgical retraction

Electrosurgery

Occasionally surgical removal of tissue and arrest of haemorrhage is required prior to taking an impression. For example, when a subgingival core is placed with a poor contour and surface finish, the gingivae can become hyperplastic, hindering impression taking following tooth preparation. Electrosurgery units utilize high-frequency electrical currents which can cut and/or coagulate soft tissue and have interchangeable active electrode tips of different configurations; probably the most commonly used are the loop and straight tip (Figure 13.19). The cutting tips are inserted into a handpiece connected to a current generator and the electrical circuit is completed to a passive electrode which is usually a plate placed beneath the patient's shoulder blade whilst lying in the dental chair (Figure 13.20). The passive electrode plate should not make direct contact with the patient's skin.

When using electrosurgery the working field should be dry and metal-free instruments should be used to reduce the risk of conduction burns. Electrosurgery around metal restorations should also be done with care so as not to



Figure 13.18 Expasyl being syringed into the gingival sulcus (left); slight blanching of the tissues can be seen as the tissues are retracted. The Expasyl is left in situ for up to 2 minutes and then rinsed off with a three-in-one syringe until completely removed.

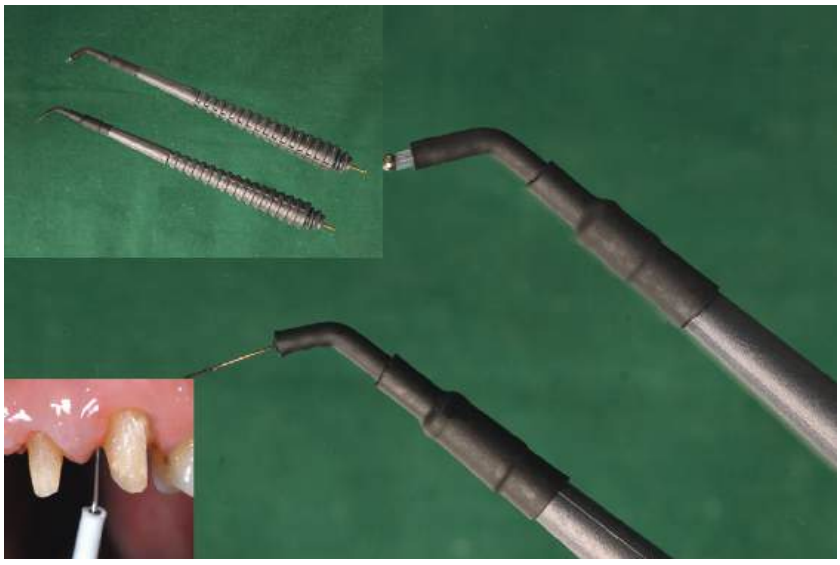


Figure 13.19 Two electrosurgery tips: a straight (left) and ball end (right). The straight tip or loop (not shown) is probably the best for gingival tissue management.



Figure 13.20 Electrosurgery unit with main box with setting dial, the electrosurgery straight tip to its right and the passive electrode plate to the far right which is placed beneath the patient's shoulder.

contact the restoration. Considerable heat is generated at the active electrode cutting tip and can cause necrosis of soft tissues and bone if used incorrectly. To reduce the heat generated, smooth sweeping movements are required; the tip should not stay stationary at any point. Cutting should not be repeated in the same area until the tissues have had time to cool which can take 5 seconds or more. Cutting movements should therefore be rehearsed prior to actual surgery. Once happy with the cut planned, the power can be applied to the unit and surgery commenced. The power may need to be increased to achieve adequate cutting of the soft tissue.

High-speed cutting burs

High-speed cutting burs have also been used to remove soft tissues surgically (Figure 13.21). Whilst quick, such 'curettage' is traumatic and associated with considerable bleeding; as such, they are rarely used.

Lasers

Lasers have also been used to cut soft tissue effectively and arrest haemorrhage; however, their use in the management of gingival tissues is uncommon. They lack tactile feedback and the equipment required is expensive. There are also health and safety issues with the use of lasers which preclude their use in the vast majority of surgeries.



Figure 13.21 An example of tissue trimming burs.

It must be remembered that some forms of gingival retraction are particularly traumatic: the greater the trauma, the greater the risk of gingival recession once the tissues have healed. The least traumatic technique should therefore be chosen in each clinical situation to bring about the desired result.

DISINFECTION OF THE IMPRESSION

It is essential for infection control that all impressions are disinfected prior to sending them to the laboratory. Impressions can be sprayed or immersed in a disinfectant. A number of disinfectants have been used previously for dental impressions and include chlorine, aldehyde, iodine or phenol-based solutions. The most commonly used are sodium hypochlorite and glutaraldehyde. Polyether and alginate can expand if soaked in a disinfectant solution for longer than 10 and 15 minutes, respectively. These materials should therefore be sprayed with an appropriate disinfectant and placed in a sealed bag or immersed for less than 10 minutes in a disinfectant solution. Addition cured silicones are dimensionally stable and can be soaked in most disinfectant solutions for the desired duration.

It is important for each material being used (impression and disinfectant) that the manufacturer's instructions are followed. It has been recommended that the minimum disinfection procedure should include immersion of the impression in a 1% sodium hypochlorite solution for at least 10 minutes.

SUMMARY

Good atraumatic tooth preparations and periodontal health make impression taking easier and reduce the need for any additional soft tissue gingival management. In general, a two-phase, single-stage impression technique in a disposable rigid stock tray and a polyether monophase impression (also in a disposable stock tray) are the most commonly used materials and techniques suitable for most fixed prosthodontics procedures. Where soft tissue gingival management is required prior to taking an impression, various options are available which are outlined in this chapter; the least traumatic should be chosen for each clinical situation. Following these guidelines will ensure good impressions, models and restorations.

FURTHER READING

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